MASS GAP OF FOUR-DIMENSIONAL LATTICE YANG–MILLS THEORY IN A FOREST GAUGE

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Abstract. *We fix the gauge of four-dimensional lattice Yang–Mills theory on a field of straight lines that meets every plaquette in exactly one link, so that each free link interacts with twelve others instead of eighteen. In this gauge the theory has a unique infinite-volume state with exponential decay of correlations and a positive mass gap at fixed lattice spacing, for SU(2) up to the limit of every single-link argument in a gauge of this kind, for every SU(N) beyond the curvature limit of the ungauged theory, and for SU(3) on a range more than twice as large as was known before.*

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1. Introduction

Wilson's lattice gauge theory [32] assigns an element Q_e of a compact group $G \subset \mathrm{U}(N)$ to every link e of $\mathbb{Z}^4$ and weighs a configuration by $\exp(N\beta \sum_p \operatorname{Re}\operatorname{Tr} Q_p)$, where Q_p is the ordered product of the four link variables around the plaquette p.[1] At small $|\beta|$ the theory is under control: it has a unique infinite-volume state, correlations decay exponentially, and the transfer matrix has a gap [27, 28]. The question is how far into the coupling range such statements can be proved with explicit constants, since the problem of the continuum limit [20] begins where strong coupling ends and Bałaban's weak-coupling analysis [3, 4, 5] takes over.

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[1] This is the normalisation of Shen, Zhu and Zhu [29] and of our earlier paper [8]. The conventional inverse coupling is $\beta_W = N^2\beta$, so the weight is $\exp(\frac{\beta_W}{N}\sum_p \operatorname{Re}\operatorname{Tr} Q_p)$; for SU(2), identified with the unit quaternions, $\frac{1}{2}\operatorname{Tr} Q_p = \operatorname{Re} Q_p$ and $\beta_W = 4\beta$. Standard accounts of the lattice theory are the books of Montvay and Münster [24] and of Gattringer and Lang [17]; the Hamiltonian version is due to Kogut and Susskind [23].

Two methods give explicit ranges. The Bakry–Émery method [1, 2] of Shen, Zhu and Zhu [29] bounds the Hessian of the action; in [8] we showed that the sharp bound is $4d = 16$ and that no curvature argument for the ungauged measure goes beyond $|\beta| < 1/32$. The Dobrushin method [13, 14] bounds the influence of each link on the conditional law of another; for SU(2) the single-link conditional law is a von Mises–Fisher law on the three-sphere, the sharp Kantorovich modulus of that family is one quarter [8, Thm. 7], and every link has eighteen partners, which gives $\beta_W < 2/9$ and nothing beyond, because the coefficients are attained.

Both limits come from the same count. A link of $\mathbb{Z}^4$ lies in six plaquettes and shares each with three other links. Gauge invariance makes part of this redundant: in a box, the variables on a forest of links can be set to the identity without changing any gauge-invariant expectation [12].[2] The forest we use is a field of straight lines with one line through every site, chosen so that every plaquette contains exactly one fixed link. Then every free link has exactly twelve free partners, two in each of its plaquettes (Lemma 2.2), and both methods improve.

Theorem 1.1 (SU(2))**.** *Let $G = \mathrm{SU}(2)$ and $0 \le \beta_W < \frac{1}{3}$. For every gauge-invariant local observable O the expectations $W_\Lambda(O)$ in boxes with free boundary converge, along every sequence of boxes that exhausts $\mathbb{Z}^4$, to a limit $\omega(O)$ that does not depend on the sequence. The state ω is invariant under translations and the hypercubic group and reflection positive; truncated correlations of Lipschitz observables decay at every rate below $-\log(3\beta_W)$; and the transfer matrix has spectral gap*

$$m \ge -\log(3\beta_W) > 0.$$

For $-\frac{1}{3} < \beta_W < 0$ all statements except reflection positivity and the gap hold.

The step behind Theorem 1.1 is short once the line field is in place.[3] The number $\frac{1}{3}$ is also the end of the road for this kind of argument.

Theorem 1.2 (Ceiling)**.** *Let F' be any set of links of $\mathbb{Z}^4$ without cycles, and give two distinct links outside F' that share a plaquette the coefficient $|\beta|$, the largest single-link Kantorovich coefficient of* SU(2)*. Then the weighted Dobrushin condition with these coefficients fails for $\beta_W \ge \frac{1}{3}$. For the line field the coefficients are attained, so no single-link Kantorovich argument in that gauge reaches $\beta_W = \frac{1}{3}$.*

The curvature method gains as well, and for every N. Let $\operatorname{Hess} S_0$ be the Hessian of $\sum_p \operatorname{Re}\operatorname{Tr} Q_p$ on G^E and F the set of fixed links.

Theorem 1.3 (Curvature in the gauge)**.** *For every N, every field Q and every tangent vector $v = (X_e Q_e)$ with $X_e = 0$ for $e \in F$,*

$$|\operatorname{Hess} S_0(v, v)| \le \tfrac{29}{2}|v|^2,$$

[2] Creutz used a maximal tree to remove the gauge freedom completely; the temporal gauge, with all links in one direction fixed, is the special case of a family of straight lines. In a maximal tree almost every link is fixed in some region and none in others, so the number of partners of a free link is not uniform, and the temporal gauge leaves fourteen partners per link. The point here is a pattern that fixes exactly one link in every plaquette.

[3] The combination of a forest gauge fixing with the sharp quarter of [8] was first assembled during a run of Anaxia, the reasoning engine of the BLLM system [6], release v3.0.5. In that run the engine combines proof steps along typed links (a step that produces a single-link Kantorovich bound may only feed a step that consumes one), and in every run with typed crossing it reached a composite accepting a lattice mass gap for four-dimensional SU(2) at $\beta_W < \frac{1}{3}$, which no untyped or random run reached. The composite was first proposed by the language model inside the engine, and the engine's network reproduces it. The engine did not supply the line field, the free-boundary limit, the ceiling, the curvature bound or anything in the SU(3) part; the authors wrote and checked every proof in this paper. Release v3.1.0 of the engine [7] records its Yang–Mills runs in the failure log: the narrowing of the target from the continuum gap to explicit couplings in this gauge, the change from analytic one-link bounds to bounds certified box by box, and the change to blocks of several links. These records follow the course of the work and supply no step of the proofs.

TABLE 1. Explicit ranges of uniqueness, decay and a transfer-matrix gap for four-dimensional lattice Yang–Mills theory. The first three rows hold for the ungauged measure; the last three are the results of this paper.

Method	SU(2), β_W	SU(3), β_W
Curvature, Shen–Zhu–Zhu [29]	$< \frac{1}{12}$	$< \frac{3}{16}$
Curvature, sharp [8]	$< \frac{1}{8}$	$< \frac{9}{32}$
Dobrushin, sharp quarter [8]	$< \frac{2}{9}$	
Curvature, forest gauge	$< \frac{4}{29}$	$< \frac{9}{29}$
Dobrushin, forest gauge	$< \frac{1}{3}$	$\leq \frac{45}{64}$

against the sharp constant 16 *without gauge fixing. Consequently, for* $G = \mathrm{SU}(N)$ *and* $|\beta| < \frac{1}{29}$*, or* $G = \mathrm{SO}(N)$*,* $N \geq 3$*, and* $|\beta| < \frac{N-2}{58N}$*, the conclusions of Theorem 1.1 hold with some rate* $c > 0$ *in place of* $-\log(3\beta_W)$*. For* SU(3) *this is* $\beta_W < \frac{9}{29} = 0.310$*.*

No Bakry–Émery argument for the ungauged measure can pass $|\beta| = \frac{1}{32}$ [8, Thm. C]; the gauge fixing changes the measure, not the metric, which is why it can. For SU(2) the curvature range $\beta_W < \frac{4}{29}$ lies inside that of Theorem 1.1. For SU(3) the Dobrushin route goes much further.

Theorem 1.4 (SU(3))**.** *Let* $G = \mathrm{SU}(3)$ *and* $0 \leq \beta_W \leq \frac{45}{64} = 0.703$*. Then the conclusions of Theorem 1.1 hold, with decay at every rate below* $-\log(1.4218\,\beta_W)$ *and a transfer-matrix gap*

$$m \geq -\log(1.4218\,\beta_W) > 0.$$

The proof of Theorem 1.4 bounds the Kantorovich modulus of the one-link measure $\nu_b(dU) \propto e^{\mathrm{Re}\,\mathrm{Tr}(Ub)}dU$ on SU(3) uniformly over staple sums b. Its analytic part is a lower bound for the spectral gap of ν_b through Bochner's formula with a cross term (Lemma 6.3), a corrector that blends the exact eigenfunction of the Haar Laplacian with the drift (Lemma 6.4), the reduction of b to diagonal matrices (Lemma 6.5), the character expansion of the moments of ν_b (Lemma 6.6) and an interpolation inequality on boxes of staple sums (Lemma 6.7). The remaining step evaluates the resulting bound on 865 boxes with rigorous error bounds; the theorem rests on that computation, which is described in full in Section 6.7 and Appendix D.

All four theorems are statements at fixed lattice spacing. They do not touch the continuum limit, the crossover to Bałaban's regime, or a gap in physical units, which together are the Clay problem; Section 7 says where each method stops and what remains of the four open problems of [8]. The obstruction proved in [9], that a lattice gap uniform in lattice units forces trivial continuum two-point functions, applies to all of them.

Section 2 constructs the line field. Section 3 shows that a Dobrushin bound for the gauge-fixed specification gives the full conclusion of Theorem 1.1. Sections 4, 5 and 6 prove Theorems 1.1–1.4, and Section 7 measures what is left. The appendices contain the residue table of the line field, the comparison lemma, the blocks of the local Hessian inequality and the details of the certified computation.

2. The line field

Write $e_0, \dots, e_3$ for the unit vectors of $\mathbb{Z}^4$ and (x, μ) for the link from x to $x + e_\mu$. A *direction field* is a map $\sigma : \mathbb{Z}^4 \to \{0, 1, 2, 3\}$; its links are $F = \{(x, \sigma(x)) : x \in \mathbb{Z}^4\}$. At each site put $\pi_x(\mu) = \sigma(x + e_\mu)$. The plaquette at x in the plane $\{\mu, \nu\}$ has the links (x, μ), $(x + e_\mu, \nu)$, $(x + e_\nu, \mu)$, (x, ν), so it contains

$$[\sigma(x) \in \{\mu, \nu\}] + [\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu] \tag{2.1}$$

links of F.

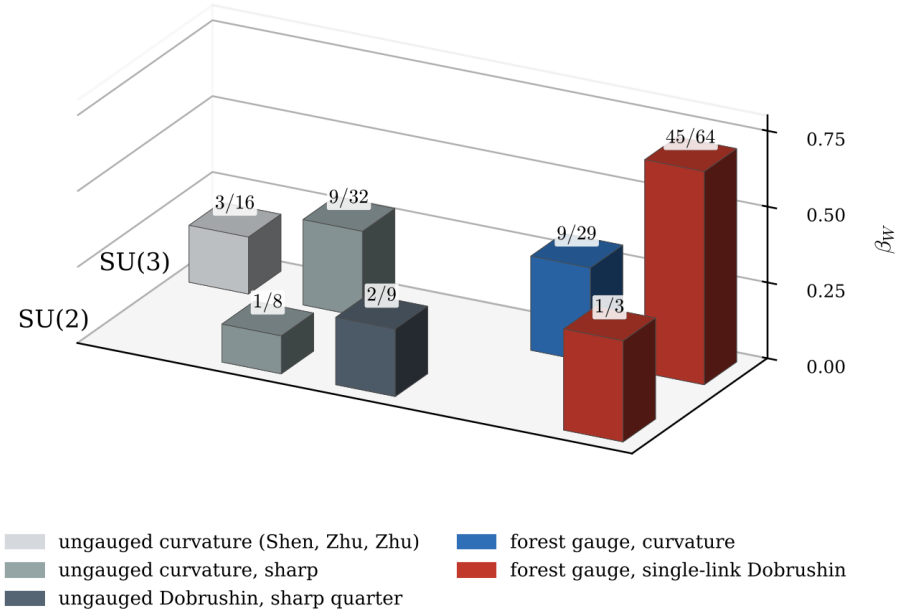


FIGURE 1. Ranges of Table 1. The height of each bar is the largest inverse coupling β_W covered by the method. Grey bars are methods for the ungauged measure; blue and red bars use the forest gauge. For SU(2) the red bar ends at the ceiling of Theorem 1.2; for SU(3) it is Theorem 1.4.

Lemma 2.1. *Every plaquette of $\mathbb{Z}^4$ contains exactly one link of F if and only if, at every site x, π_x fixes $\sigma(x)$ and permutes the other three directions cyclically.*

Proof. Fix x, let $s = \sigma(x)$ and let R be the set of the other three directions. For a plane $\{s, \nu\}$ the count (2.1) is $1 + [\pi_x(s) = \nu] + [\pi_x(\nu) = s]$; it equals one for all $\nu \in R$ exactly when $\pi_x(s) = s$ and $\pi_x(\nu) \neq s$ for $\nu \in R$, that is, when π_x fixes s and maps R into R. For a plane $\{\mu, \nu\} \subset R$ the count is $[\pi_x(\mu) = \nu] + [\pi_x(\nu) = \mu]$. Draw an arrow $\mu \to \pi_x(\mu)$ for every $\mu \in R$ with $\pi_x(\mu) \in R \setminus \{\mu\}$. Each vertex of R has at most one outgoing arrow, and the condition says that each of the three pairs of R carries exactly one arrow. Three arrows on three vertices with out-degree at most one give out-degree exactly one everywhere, and a tournament on three vertices with all out-degrees one is a directed three-cycle. □

From now on σ is the field of period two given in Table 2 of Appendix A, read from x modulo two. The table lists π_x at each of the sixteen residue classes; in each row π_x fixes $\sigma(x)$ and is a three-cycle on the rest, so every plaquette contains exactly one link of F by Lemma 2.1.[4]

Lemma 2.2. (i) *F is a disjoint union of bi-infinite straight lines, one through every site: every site has exactly one outgoing and one incoming link of F, and both point in the direction $\sigma(x)$.*

(ii) *For every box Λ (a product of four finite intervals, with the links and plaquettes it contains), $F \cap E(\Lambda)$ is a disjoint union of segments, hence a forest.*

(iii) *A link $e = (x, \mu) \notin F$ lies in six plaquettes. Four of them contain a link of F that shares an end point with e, and these lie in the planes $\{\mu, \sigma(x)\}$ and $\{\mu, \pi_x(\mu)\}$. In the other two, which lie in the remaining plane $\{\mu, \rho\}$, the link of F is opposite to e.*

[4]There are exactly eight period-two direction fields with this property, all equivalent to σ under the symmetries of the lattice. A field with one fixed link at every site and one in every plaquette exists only in four dimensions: a link lies in $2(d-1)$ plaquettes and a site carries $d(d-1)/2$ of them, and $2(d-1) = d(d-1)/2$ forces $d = 4$. The statements of this section are also checked by `decide` in the Lean file `GaugeFix.lean` of the archive.

(a) the line field in the slice $x_3 = 0$ (b) one plaquette in the gauge

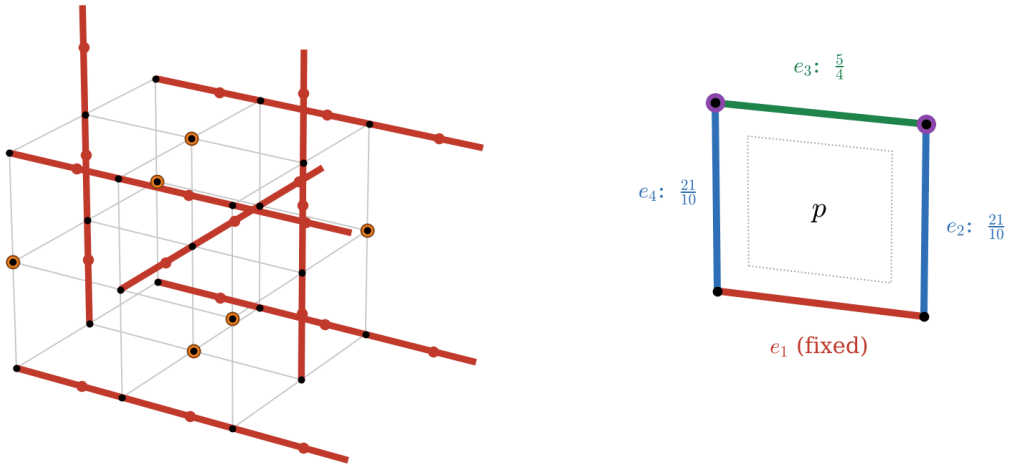


FIGURE 2. (a) The line field in the slice $x_3 = 0$ of a box of side three. Thick red segments are the links of F in the directions $0, 1, 2$, with a dot marking the direction of travel; an orange ring marks a site whose line leaves the slice in direction 3. Every line is straight (Lemma 2.2(i)), and the segments that leave the box are its boundary stubs. (b) One plaquette with its fixed link e_1. The weights are those of the local inequality of Lemma 5.2 for the three free links: $\frac{21}{10}$ on the two links adjacent to e_1 and $\frac{5}{4}$ on the opposite one. The two purple corners are the ones that enter the corner form (5.3) once $Y_1 = 0$.

(iv) *Every link outside F shares a plaquette with exactly twelve other links outside F, two in each of its plaquettes.*

Proof. (i) Since π_x fixes $\sigma(x)$, $\sigma(x+e_{\sigma(x)}) = \sigma(x)$. Because σ has period two, $x-e_\mu \equiv x+e_\mu$, so also $\sigma(x-e_{\sigma(x)}) = \sigma(x)$, and the link $(x-e_{\sigma(x)}, \sigma(x))$ is in F. Conversely, if $(x-e_\nu, \nu) \in F$, then $\sigma(x-e_\nu) = \nu$, and straightness at $x-e_\nu$ gives $\sigma(x) = \nu$. So the incoming link is unique and continues the outgoing one.

(ii) A line meets a box in a segment, and distinct lines are disjoint.

(iii) The plaquettes of e are, for each $\nu \neq \mu$, the one at x and the one at $x-e_\nu$ in the plane $\{\mu, \nu\}$. A link of F sharing an end point with e is the outgoing or the incoming link of F at x or at $x+e_\mu$. At x both point in the direction $s = \sigma(x) \neq \mu$, and they lie in the plaquettes at x and at $x-e_s$ in the plane $\{\mu, s\}$. At $x+e_\mu$ both point in the direction $t = \pi_x(\mu)$, which lies in $R \setminus \{\mu\}$ because π_x is a three-cycle on R, and they lie in the plaquettes at x and at $x-e_t$ in the plane $\{\mu, t\}$. These four plaquettes are distinct because $s \neq t$. Each plaquette contains exactly one link of F, so in the two plaquettes of the third plane $\{\mu, \rho\}$ the link of F is a link of direction μ or ρ not touching e; the links of direction ρ in such a plaquette touch e, so it is the opposite link.

(iv) Each of the six plaquettes contains e, one link of F and two other links, and two distinct links share at most one plaquette. □

3. GAUGE FIXING AND THE FREE-BOUNDARY STATE

Let $G \subset \mathrm{U}(N)$ be compact with Haar measure dU, and let Λ be a box. The Wilson measure with free boundary is

$$W_\Lambda(dQ) = \frac{1}{Z_\Lambda} \exp\Big(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re}\operatorname{Tr} Q_p\Big) \prod_{e \in E(\Lambda)} dQ_e.$$

The measure is invariant under the gauge transformations $Q_{(x,y)} \mapsto g_x Q_{(x,y)} g_y^{-1}$ with $g \in G^\Lambda$. Call the links outside F *active* and write A for their set. For a function O of the links, O^F is O with every link of F set to the identity.

Lemma 3.1 (Gauge fixing)**.** *Let μ_Λ^F be the probability measure on $G^{A \cap E(\Lambda)}$ whose density with respect to Haar measure is proportional to $\exp(N\beta \sum_{p \in P(\Lambda)} \operatorname{Re} \operatorname{Tr} Q_p^F)$. For every gauge-invariant O on $G^{E(\Lambda)}$,*

$$W_\Lambda(O) = \mu_\Lambda^F(O^F).$$

Proof. Each segment of $F \cap E(\Lambda)$ is a path $x_0, x_1, \dots, x_k$ along one direction. Given the variables on F, put $g_{x_0} = 1$ and $g_{x_{i+1}} = g_{x_i} Q_{(x_i, x_{i+1})}$ on every segment. The transformed field has $Q_e^g = 1$ on $F \cap E(\Lambda)$, and for fixed values on F the map $Q_A \mapsto Q_A^g$ is a product of left and right translations, which preserves Haar measure. Gauge invariance of the action and of O and the change of variables give the claim. □

No leftover holonomy appears because $F \cap E(\Lambda)$ is a forest. The measures μ_Λ^F are finite-volume versions of the specification γ^F on G^A: for $e \in A$, γ_e^F resamples Q_e from the law with density proportional to $\exp(N\beta \sum_{p \ni e} \operatorname{Re} \operatorname{Tr} Q_p^F)$ given all other active links. No infinite-volume gauge transformation is performed; γ^F is a specification in its own right. Fix a metric δ on G of diameter D that is bi-invariant (for SU(2) the chordal metric on S^3, $D = 2$; for SU(N) the Hilbert–Schmidt distance). The Kantorovich coefficient of e' on e is

$$C_{ee'} = \sup \frac{W_1(\gamma_e^F(\cdot \mid Q), \gamma_e^F(\cdot \mid Q'))}{\delta(Q_{e'}, Q'_{e'})},$$

the supremum over pairs of configurations that differ only at e'. It vanishes unless e and e' share a plaquette.

Proposition 3.2. *Suppose that every row and every column of $(C_{ee'})$ sums to at most $\alpha < 1$. Then:*

(i) *γ^F has exactly one Gibbs measure μ_F;*
(ii) *for every gauge-invariant local O the limit $\omega(O) = \lim W_\Lambda(O)$ exists along every sequence of boxes exhausting $\mathbb{Z}^4$ and equals $\mu_F(O^F)$;*
(iii) *ω is invariant under translations and the hypercubic group;*
(iv) *if $\beta \geq 0$, ω is reflection positive for reflections in hyperplanes through sites and in hyperplanes bisecting links;*
(v) *for gauge-invariant local O, O' with O Lipschitz in each link, $|\operatorname{Cov}_\omega(O, O')| \leq C_{O,O'} e^{-\lambda \operatorname{dist}(O,O')}$ for every $\lambda < -\log \alpha$;*
(vi) *if $\beta \geq 0$, the transfer matrix of ω has spectral gap at least $-\log \alpha$.*

Proof. (i) Gibbs measures exist by compactness, and the comparison lemma of Appendix B with $S = \emptyset$ shows that any two coincide.

(ii) Extend μ_Λ^F by Haar measure on the active links outside Λ, and let S be the set of active links with a plaquette outside $P(\Lambda)$; this includes every active link outside $E(\Lambda)$. For an active link $e \notin S$ all six plaquettes of e lie in $P(\Lambda)$, so the conditional law of Q_e under the extended measure is γ_e^F. Let T be the support of O^F and $\lambda < -\log \alpha$. The comparison lemma with weights $e^{-\lambda \operatorname{dist}(\cdot, T)}$ gives

$$|\mu_\Lambda^F(O^F) - \mu_F(O^F)| \leq L \frac{\theta D}{1 - \theta} \sum_{z \in S} e^{-\lambda \operatorname{dist}(z,T)},$$

with $\theta = \alpha e^\lambda < 1$ and L a Lipschitz constant of O^F; the sum is at most $C_T R^3 e^{-\lambda R}$ when R is the distance from T to the complement of Λ. For O continuous but not Lipschitz, approximate uniformly. Lemma 3.1 gives $W_\Lambda(O) = \mu_\Lambda^F(O^F)$.

(iii) $W_\Lambda(\tau_v O) = W_{\Lambda - v}(O)$, and the boxes $\Lambda - v$ also exhaust $\mathbb{Z}^4$; by (ii) the limit does not depend on the sequence. The line field is a device for computing W_Λ and is never shifted. The same argument applies to the hypercubic group.

(iv) For $\beta \geq 0$ the measure W_Λ is reflection positive for every box symmetric under the reflection [27], and reflection positivity passes to the limit along such boxes, which is ω by (ii).

(v) $\mathrm{Cov}_\omega(O, O') = \mathrm{Cov}_{\mu_F}(O^F, O'^F)$, and setting the links of F to the identity keeps the Lipschitz property. The bound is Corollary B.2.

(vi) Let ϑ be a time reflection and τ_t a time translation. For gauge-invariant local O that are Lipschitz in each link, (v) gives $|\omega(\vartheta(\bar{O})\tau_t O) - |\omega(O)|^2| \leq C_O e^{-\lambda t}$. On the support of O these functions are dense in the gauge-invariant part of $L^2(\omega)$: gauge-invariant functions are functions on the compact quotient of G^E by the gauge group, Lipschitz functions for the quotient metric are dense in L^2 of every finite measure on it, and they lift to gauge-invariant Lipschitz functions. By [8, Prop. 23] the spectrum of the transfer matrix on the complement of the vacuum lies in $[0, e^{-\lambda}]$, for every $\lambda < -\log \alpha$. □

The proposition replaces the torus limits and the uniqueness of the ungauged Gibbs measure used in [8, Thm. F] by the free-boundary limit, which has all the properties the reconstruction of the transfer matrix needs.[5]

4. SU(2) up to one third

Proof of Theorem 1.1. Identify SU(2) with the unit quaternions and use the chordal metric. For an active link e and a plaquette $p \ni e$, the staple s_p is the product of the other three link variables in the order that makes $\mathrm{Re}(Q_e s_p) = \frac{1}{2}\mathrm{Tr}\, Q_p$, with links of F equal to one. Since $N\beta\,\mathrm{Re}\,\mathrm{Tr}\, Q_p = 4\beta\langle Q_e, \bar{s}_p\rangle$ in $\mathbb{R}^4$, the conditional law of Q_e is the von Mises–Fisher law $\nu_b(dx) \propto e^{\langle x,b\rangle}dx$ on S^3 with $b = 4\beta\sum_{p\ni e}\bar{s}_p$. Changing a partner e' by δ in the chordal metric changes one staple by an isometric image of that displacement, so b moves by $4|\beta|\delta$, and the sharp quarter [8, Thm. 7], $W_1(\nu_b, \nu_{b'}) \leq \frac{1}{4}|b - b'|$, gives $C_{ee'} \leq |\beta|$. By Lemma 2.2(iv) every row and column has twelve nonzero entries, so $\alpha \leq 12|\beta| = 3|\beta_W|$. Proposition 3.2 applies for $|\beta_W| < \frac{1}{3}$. □

Proof of Theorem 1.2. For the line field, every staple of an active link contains two active links, so the six staples can be chosen with sum zero while the staple through a given partner e' moves freely. At $b = 0$ the derivative of ν_b in the direction h has Kantorovich norm $\frac{1}{4}|h|$ [8, Lemma 6(i)], so $C_{ee'} = |\beta|$ exactly, and every row sum is $12|\beta| = 3|\beta_W|$.

Now let F' be any forest and C' the matrix of the statement. Suppose a positive weight w satisfies $C'w \leq \alpha w$. For a box Λ_n of side n let C_n be the principal submatrix on the active links of $E(\Lambda_n)$; then $C_n w \leq \alpha w$ on these links, and the Collatz–Wielandt inequality for nonnegative matrices gives $\rho(C_n) \leq \alpha$. Since C_n is symmetric, $\rho(C_n)$ is at least the mean row sum. The box has $4n^4 - O(n^3)$ links and n^4 sites, and a forest has fewer links than sites, so the number a of active links is at least $3n^4 - O(n^3)$. Let a_p be the number of active links of the plaquette p; the box has $P = 6n^4 - O(n^3)$ plaquettes and $\sum_p a_p = 6a - O(n^3)$. Two links share at most one plaquette, so the row sums add up to $|\beta|\sum_p a_p(a_p - 1)$, and by convexity

$$\frac{|\beta|}{a}\sum_p a_p(a_p - 1) \geq \frac{|\beta|P}{a}\,\bar{a}(\bar{a} - 1) = 6|\beta|(\bar{a} - 1) - o(1),$$

with $\bar{a} = \sum_p a_p/P = 6a/P - o(1) \geq 3 - o(1)$. Hence $\alpha \geq 12|\beta| - o(1)$ for every n, and $\alpha \geq 12|\beta| = 3|\beta_W|$. □

Without gauge fixing the same count gives eighteen partners and the ceiling $\beta_W = \frac{2}{9}$ of [8, Prop. 13]; the temporal gauge, a family of lines in one direction, leaves fourteen. The line field is optimal among forests: twelve is the mean number of partners forced by the number of free links, and it is attained at every link.

[5] On a torus the line field closes up into cycles, and the holonomies along them cannot be removed by a gauge transformation; the argument never uses a torus. The state ω is defined only on gauge-invariant observables, which is all that the Osterwalder–Schrader reconstruction [25, 26, 27] uses.

5. Curvature in the gauge

In this section $G = \mathrm{SU}(N)$, $\mathrm{SO}(N)$ or $\mathrm{U}(N)$ with the inner product $\langle X, Y\rangle = -\operatorname{Tr}(XY)$ on the Lie algebra, so that $\mathrm{Ric}_{\mathrm{SU}(N)} = \frac{N}{2}$ and $\mathrm{Ric}_{\mathrm{SO}(N)} = \frac{N-2}{4}$. A tangent vector at a field Q is $v = (X_e Q_e)$, $|v|^2 = \sum_e |X_e|^2$. For the plaquette at x in the plane $\mu < \nu$ number its links $e_1 = (x, \mu)$, $e_2 = (x + e_\mu, \nu)$, $e_3 = (x + e_\nu, \mu)$, $e_4 = (x, \nu)$, transport their vectors to x,

$$Y_1 = X_{e_1}, \quad Y_2 = Q_{e_1} X_{e_2} Q_{e_1}^*, \quad Y_3 = Q_{e_4} X_{e_3} Q_{e_4}^*, \quad Y_4 = X_{e_4}, \tag{5.1}$$

and write $W = Q_p$. The second derivative of $\operatorname{Re}\operatorname{Tr} Q_p$ along the geodesic $Q_e(t) = e^{tX_e} Q_e$ is [8, Eq. (5)]

$$\begin{aligned} f_p = {} & \sum_{i=1}^{4} \operatorname{Re}\operatorname{Tr}(Y_i^2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_1 Y_2 W) + 2\operatorname{Re}\operatorname{Tr}(Y_3 Y_4 W) \\ & - 2 \sum_{i\in\{1,2\},\, j\in\{3,4\}} \operatorname{Re}\operatorname{Tr}(Y_i W Y_j), \end{aligned} \tag{5.2}$$

so that $\operatorname{Hess} S_0(v, v) = \sum_p f_p$. The corner form of p is

$$c_p = 2\langle Y_1, Y_4\rangle - 2\langle Y_1, Y_2\rangle + 2\langle Y_2, W Y_3 W^*\rangle - 2\langle Y_3, Y_4\rangle. \tag{5.3}$$

It collects the inner products of the transported vectors of two links that meet at a right angle at a corner of p.

Lemma 5.1. *On $\mathbb{Z}^4$, or on a box with free boundary, for every field Q and every square-summable X,*

$$\begin{aligned} \sum_y \Big[|D_y X|^2 + \sum_\mu \big| X_{y,\mu} + Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu} \big|^2 \Big] \\ = 4|X|^2 + \sum_p c_p, \end{aligned}$$

where $D_y X = \sum_\mu [X_{y,\mu} - Q^*_{y-e_\mu,\mu} X_{y-e_\mu,\mu} Q_{y-e_\mu,\mu}]$ is the covariant divergence and terms with a link outside the box are omitted.

Proof. At a site y write o_μ for the vector of the outgoing link in direction μ and i_μ for the transported vector of the incoming one, so that $D_y X = \sum_\mu (o_\mu - i_\mu)$. Every link is outgoing at one end and incoming at the other, so the squares $|o_\mu|^2 + |i_\mu|^2$, which appear once in $|D_y X|^2$ and once in $|o_\mu + i_\mu|^2$, add up to $4|X|^2$. The collinear cross term $-2\langle o_\mu, i_\mu\rangle$ of $|D_y X|^2$ cancels the term $+2\langle o_\mu, i_\mu\rangle$ of $|o_\mu + i_\mu|^2$. The remaining cross terms of $|D_y X|^2$ pair two perpendicular ends at y, and each such pair is a corner of exactly one plaquette, with the sign and the transport of (5.3) (this is the proof of [8, Lemma 1]). In a box two perpendicular ends at y span a plaquette of the box, because a box is a product of intervals. □

Lemma 5.2. *Let $k \in \{1, 2, 3, 4\}$, let $k \pm 1$ (mod 4) be the positions adjacent to k and $k + 2$ the opposite one. For every N, every $W \in \mathrm{U}(N)$ and all $Y_1, \dots, Y_4 \in \mathfrak{u}(N)$ with $Y_k = 0$,*

$$\pm f_p + \tfrac{9}{10} c_p \le \tfrac{21}{10}(|Y_{k-1}|^2 + |Y_{k+1}|^2) + \tfrac{5}{4}|Y_{k+2}|^2.$$

Proof. Replacing W by $-W$ changes the sign of f_p and fixes c_p, so the sign + suffices. Conjugation by a unitary matrix preserves both sides, so we may take $W = \operatorname{diag}(w_a)$. The form $\mathcal{F} = \mathrm{RHS} - f_p - \frac{9}{10} c_p$ then splits into a block for each diagonal position a, in the real variables $y_i = -i(Y_i)_{aa}$, and a block for each pair $a \neq b$, in the complex variables $z_i = (Y_i)_{ab}$, since $(Y_i)_{ba} = -\bar{z}_i$. Appendix C shows that every block is positive definite. □

Proof of the bound in Theorem 1.3. Every plaquette has exactly one link of F, and X vanishes on it. By Lemma 2.2(iii) each active link is adjacent to the fixed link in four of its

plaquettes and opposite to it in two. Summing Lemma 5.2 over the plaquettes,

$$\sum_p (\pm f_p + \tfrac{9}{10} c_p) \le (4 \cdot \tfrac{21}{10} + 2 \cdot \tfrac{5}{4}) |X|^2 = \tfrac{109}{10} |X|^2.$$

By Lemma 5.1, $\frac{9}{10} \sum_p c_p = \frac{9}{10}\Sigma - \frac{36}{10}|X|^2$, where $\Sigma \ge 0$ is the left side of the lemma. Hence

$$\pm \operatorname{Hess} S_0(v,v) + \tfrac{9}{10}\Sigma \le (\tfrac{109}{10} + \tfrac{36}{10})|X|^2 = \tfrac{29}{2}|X|^2.$$

In a box, links near the boundary lie in fewer plaquettes and receive less. □

The factor $\frac{9}{10}$ is chosen so that the corners are paid by the divergence identity at the price $\frac{36}{10}$ while the local blocks stay positive; Lemma 5.1 is the divergence identity of [8, Lemma 1] with the collinear term written as a square, so nothing is lost by Cauchy–Schwarz.

Corollary 5.3. *Let $G = \mathrm{SU}(N)$ and $|\beta| < \frac{1}{29}$, or $G = \mathrm{SO}(N)$, $N \ge 3$, and $|\beta| < \frac{N-2}{58N}$. For every box the measure μ_Λ^F satisfies the Bakry–Émery condition with $K = \mathrm{Ric}_G - \frac{29}{2} N|\beta| > 0$, uniformly in the box, and hence a logarithmic Sobolev inequality with constant $2/K$ and a Poincaré inequality with constant $1/K$.*

Proof. The set $\{Q_F = 1\}$ is a product factor, so its Ricci curvature is Ric_G on every link, and the Hessian of the restricted action is the restriction of $N\beta \operatorname{Hess} S_0$ to $X_F = 0$. Apply the bound and [2, Cor. 5.7.2]. □

Proof of the rest of Theorem 1.3. Decay in boxes. μ_Λ^F is the Gibbs measure of a nearest-neighbour interaction on a product of copies of G. The commutator estimate of [29, Section 4] holds with the same constants, because fixing links only removes terms, and the Poincaré constant is $1/K$, so the argument of [29, Cor. 4.11 and Rem. 4.12] gives $|\operatorname{Cov}_{\mu_\Lambda^F}(f,g)| \le c_1 e^{-c \operatorname{dist}(f,g)} (\|f\|\,\|g\| + |f|_2 |g|_2)$ for smooth local f, g, uniformly in the box, in the norms of that paper.

Limit. For boxes $\Lambda \subset \Lambda'$ let $S_s = S_\Lambda + s(S_{\Lambda'} - S_\Lambda)$ on the active links of Λ', $s \in [0,1]$, and μ_s the corresponding measure. Its Hessian is a convex combination of two Hessians, each bounded by $\frac{29}{2} N|\beta|$, so μ_s has the same constant K and the same commutator constants. Since $\frac{d}{ds}\mu_s(f) = \sum_{p \in P(\Lambda') \setminus P(\Lambda)} \operatorname{Cov}_{\mu_s}(f, N\beta \operatorname{Re}\operatorname{Tr} Q_p^F)$,

$$|\mu_{\Lambda'}^F(f) - \mu_\Lambda^F(f)| \le C_f R^3 e^{-cR},$$

R the distance from the support of f to the complement of Λ. The limit therefore exists along every increasing sequence of boxes, and two sequences are compared through their intersections, which are boxes.

Properties. Invariance, reflection positivity and decay follow as in Proposition 3.2(iii)–(v), and Proposition 23 of [8], applied to smooth gauge-invariant cylinder functions, gives a gap at least c. □

Remark 5.4. Let h_F be the supremum of $\pm \operatorname{Hess} S_0$ on the gauge-fixed directions, so that uniform curvature reaches $|\beta| < 1/(2h_F)$ and no further. At the identity field $-\operatorname{Hess} S_0$ restricted to $X_F = 0$ has the eigenvalue $9 + \sqrt{17} = 13.123$ on the torus of side two, so $9 + \sqrt{17} \le h_F \le \frac{29}{2}$. A gradient search over SU(2) fields of periods two, four and six gives $h_F = 13.97401$, attained by an abelian field; certificates made of plaquette forms and vertex forms on the period-two cell cannot go below 14.384. With this gauge the curvature method stops between $|\beta| = \frac{1}{29}$ and $\frac{1}{26.25}$.

6. SU(3) up to 45/64

In this section $G = \mathrm{SU}(3)$, $\beta \ge 0$ and $\beta_W = 9\beta$. The distance on SU(3) is $d(U,V) = \|U - V\|_{\mathrm{HS}}$, which is at most the geodesic distance for $\langle X, Y\rangle = -\operatorname{Tr}(XY)$; $K = \operatorname{conv} \mathrm{SU}(3)$; $A(M) = \frac{1}{2}(M - M^*)$ and P removes the trace. With this inner product $\mathrm{Ric} = \frac{3}{2}$, the Casimir of the fundamental representation is $\lambda_F = \frac{8}{3}$ and that of the adjoint representation is 6.

6.1. **Reduction to one link.** The conditional law of an active link is

$$\nu_b(dU) \propto e^{\operatorname{Re}\operatorname{Tr}(Ub)}dU, \qquad b = N\beta\sum_{p\ni e} S_p,$$

where S_p is the staple of e in p, a product of SU(3) matrices and identities, or its adjoint. Six staples give $b \in rK$ with $r = 6N\beta = 2\beta_W$. For a matrix h put $\varphi_h(U) = \operatorname{Re}\operatorname{Tr}(Uh)$ and

$$L_h(b) = \sup\{\operatorname{Cov}_{\nu_b}(g,\varphi_h) : g \text{ 1-Lipschitz for } d\},$$

$$\Lambda(r) = \sup_{b\in rK}\ \sup_{\|h\|_{\mathrm{HS}}=1} L_h(b).$$

Changing a partner e' changes one staple by a matrix of norm $d(Q_{e'}, Q'_{e'})$. Along the segment b_t from b to b', which stays in rK, $\frac{d}{dt}\nu_{b_t}(g) = \operatorname{Cov}_{\nu_{b_t}}(g, \operatorname{Re}\operatorname{Tr}(U(b'-b)))$, so $W_1(\nu_b,\nu_{b'}) \le N\beta\Lambda(r)\, d(Q_{e'}, Q'_{e'})$. With twelve partners per link,

$$\alpha \le 12N\beta\Lambda(r) = 2r\Lambda(r) = 4\beta_W\Lambda(2\beta_W). \tag{6.1}$$

Since K is convex and contains 0, the mean of $I, \omega I, \omega^2 I$ with $\omega^3 = 1$, $r'K \subset rK$ for $r' \le r$ and Λ is nondecreasing.

6.2. **Gradients and one-forms.**

Lemma 6.1. *In the left trivialisation, $\nabla\varphi_c(U) = -PA(cU)$; if $V = \varphi_b$ then* $\operatorname{Hess} V(X,X) = \operatorname{Re}\operatorname{Tr}(UX^2b)$*; and $\Delta\varphi_c = -\lambda_F\varphi_c$.*

Proof. $\frac{d}{dt}\operatorname{Re}\operatorname{Tr}(Ue^{tX}c)|_{t=0} = \operatorname{Re}\operatorname{Tr}(cUX) = -\langle PA(cU), X\rangle$ for $X \in \mathfrak{su}(3)$. On a group with a bi-invariant metric $\nabla_X X = 0$ for left-invariant X, so the Hessian is the second derivative along e^{tX}. The function φ_c is a matrix coefficient of the fundamental representation, on which the Casimir acts by λ_F. □

Lemma 6.2. *The bottom of the spectrum of the rough Laplacian $\nabla^*\nabla$ on L^2 one-forms of* SU(3) *is $\mu_0 = \frac{7}{6}$.*

Proof. Trivialise one-forms on the left, so that a one-form is a function $f : \mathrm{SU}(3) \to \mathfrak{su}(3)$. For an orthonormal basis X_a acting as left-invariant vector fields, $\nabla_{X_a}Y = \frac{1}{2}[X_a, Y]$ and $\nabla_{X_a}X_a = 0$, so $\nabla^*\nabla = -\sum_a(\rho(X_a) + \frac{1}{2}\operatorname{ad}X_a)^2$. On the isotypic part $V_\pi \otimes V_\pi^* \otimes \mathfrak{su}(3)$ of $L^2 \otimes \mathfrak{su}(3)$ put $A_a = \rho_\pi(X_a)$, $B_a = \operatorname{ad}X_a$ and $T_a = A_a + B_a$, the action on the tensor product. Since A_a and B_a commute, $A_a + \frac{1}{2}B_a = \frac{1}{2}(A_a + T_a)$ and $A_aT_a + T_aA_a = A_a^2 + T_a^2 - B_a^2$, so on a component $\sigma \subset \pi \otimes \mathrm{ad}$

$$\nabla^*\nabla = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{1}{4}C_{\mathrm{ad}} = \tfrac{1}{2}(C_\pi + C_\sigma) - \tfrac{3}{2},$$

with $C(p,q) = \frac{2}{3}(p^2 + q^2 + pq + 3p + 3q)$ for the representation of highest weight (p,q) [16]. The components of $(p,q)\otimes(1,1)$ are $(p+1,q+1)$, $(p+2,q-1)$, $(p-1,q+2)$, $(p+1,q-2)$, $(p-2,q+1)$, $(p-1,q-1)$ and (p,q), the last with multiplicity two, one or zero as both, one or none of p,q are positive, and only those with nonnegative labels occur. For $\pi = (0,0)$ the only component is $(1,1)$, with value $\frac{3}{2}$. For $\pi = (1,0)$ the components are $(2,1)$, $(0,2)$, $(1,0)$, with values $\frac{31}{6}$, $\frac{19}{6}$, $\frac{7}{6}$, and $(0,1)$ is conjugate. For $p+q \ge 2$ the smallest values occur at $(p-1,q-1)$, (p,q), $(p-2,q+1)$ or $(p+1,q-2)$, and each is at least $\frac{1}{3}(2(p^2+q^2+pq) + 3(p+q) - 3) - \frac{3}{2} \ge \frac{3}{2}$. □

6.3. **The gap of the one-link measure.**

Lemma 6.3. *For $b \in rK$ the spectral gap of $L = \Delta + \langle\nabla V, \nabla\cdot\rangle$, $V = \varphi_b$, on $L^2(\nu_b)$ is at least*

$$\gamma(r) = \tfrac{39}{64}\mu_0 + \tfrac{3}{2} - \tfrac{r}{2} = \tfrac{283}{128} - \tfrac{r}{2}.$$

Proof. Let $-Lf = \lambda f$ with $\lambda > 0$, $\omega = df$ and $\eta = e^{V/2}\omega$; norms are in $L^2(dU)$. Bochner's formula integrated against ν_b gives $\lambda\nu_b|\omega|^2 = \nu_b|\nabla\omega|^2 + \nu_b((\mathrm{Ric} - \mathrm{Hess}\, V)(\omega,\omega))$, and since $e^{V/2}\nabla\omega = \nabla\eta - \frac{1}{2}dV\otimes\eta$,

$$\lambda\|\eta\|^2 = \|\nabla\eta - \tfrac{1}{2}dV\otimes\eta\|^2 + \int (\mathrm{Ric} - \mathrm{Hess}\, V)(\eta,\eta). \tag{6.2}$$

Put $\mathcal{A} = \nabla\eta$ and $\mathcal{B} = \frac{1}{2}dV\otimes\eta$. Then

$$\langle\mathcal{A},\mathcal{B}\rangle = \tfrac{1}{4}\int\langle\nabla V, \nabla|\eta|^2\rangle = \tfrac{\lambda_F}{4}\int V|\eta|^2,$$

because $-\Delta V = \lambda_F V$. Writing $\|\mathcal{A}-\mathcal{B}\|^2 = \|\frac{5}{8}\mathcal{A}-\mathcal{B}\|^2 + \frac{39}{64}\|\mathcal{A}\|^2 - \frac{3}{4}\langle\mathcal{A},\mathcal{B}\rangle$ and dropping the first term,

$$\|\mathcal{A}-\mathcal{B}\|^2 \ge \tfrac{39}{64}\mu_0\|\eta\|^2 - \tfrac{1}{2}\int V|\eta|^2$$

by Lemma 6.2. With $\mathrm{Ric} = \frac{3}{2}$, (6.2) gives $\lambda\|\eta\|^2 \ge (\frac{39}{64}\mu_0 + \frac{3}{2})\|\eta\|^2 - \int(\mathrm{Hess}\, V(\eta,\eta) + \frac{1}{2}V|\eta|^2)$, and the claim follows from the pointwise bound

$$\mathrm{Re}\,\mathrm{Tr}(MX^2) + \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\, M \le \tfrac{r}{2}|X|^2, \qquad M = bU \in rK, \tag{6.3}$$

since $\mathrm{Hess}\, V(X,X) = \mathrm{Re}\,\mathrm{Tr}(MX^2)$ and $V = \mathrm{Re}\,\mathrm{Tr}\, M$. The maximum over $|X| = 1$ of the left side is convex in M, so it suffices to take $M = rW$ with $W \in \mathrm{SU}(3)$ and, after a conjugation, $W = \mathrm{diag}(e^{i\theta_k})$ with $\sum_k \theta_k = 0$; put $c_k = \cos\theta_k$. Write $X = iH$ with H Hermitian, traceless, $\mathrm{Tr}\, H^2 = 1$, with diagonal d_k and entries H_{jk}. Then

$$\mathrm{Re}\,\mathrm{Tr}(WX^2) = -\sum_k c_k d_k^2 - \sum_{j<k}(c_j + c_k)|H_{jk}|^2,$$

with $\sum_k d_k^2 + 2\sum_{j<k}|H_{jk}|^2 = 1$, so the maximum is the larger of $q_{\mathrm{off}} = \max_{j<k} -\frac{1}{2}(c_j + c_k)$ and $q_{\mathrm{diag}} = \max\{-\sum c_k d_k^2 : \sum d_k = 0,\ \sum d_k^2 = 1\}$. If l is the third index, $q_{\mathrm{off}} + \frac{1}{2}\sum_k c_k = \frac{1}{2}c_l \le \frac{1}{2}$. The value q_{diag} is minus the smaller eigenvalue of $\mathrm{diag}(c)$ compressed to the plane $\sum d_k = 0$; the eigenvalues of the compression solve $\sum_k (c_k - \mu)^{-1} = 0$, that is $3\mu^2 - 2e_1\mu + e_2 = 0$ with e_1, e_2 the elementary symmetric functions of the c_k, so $q_{\mathrm{diag}} = \frac{1}{3}(\sqrt{e_1^2 - 3e_2} - e_1)$. The inequality $q_{\mathrm{diag}} + \frac{1}{2}e_1 \le \frac{1}{2}$ is $\sqrt{e_1^2 - 3e_2} \le \frac{1}{2}(3 - e_1)$, that is $e_1^2 - 4e_2 + 2e_1 - 3 \le 0$. When $\sum\theta_k = 0$, $\sum c_k^2 = 1 + 2c_1c_2c_3$, and with this identity the left side equals $-2(1-c_1)(1-c_2)(1-c_3) \le 0$. □

The weight $\frac{5}{8}$ is the one that gives the coefficient $\frac{1}{2}$ of V, and (6.3) is an equality at $W = I$. At $r = \frac{45}{32}$, $\gamma = \frac{193}{128}$.[6]

6.4. **A blended corrector.**

Lemma 6.4. *For $b \in rK$, $\|h\| = 1$ and $s \ge 0$, put $\Gamma = \langle\nabla V, \nabla\varphi_h\rangle$ and $X_s = (1 - s\lambda_F)\varphi_h + s\Gamma$. Then*

$$L_h(b) \le s\sqrt{\nu_b|\nabla\varphi_h|^2} + \sqrt{\mathrm{Var}_{\nu_b}(X_s)/\gamma(r)}.$$

Proof. Let $-L\psi = \varphi_h - \nu_b\varphi_h$. For a 1-Lipschitz g, $|\nabla g| \le 1$ almost everywhere, because d is at most the geodesic distance, so $\mathrm{Cov}(g,\varphi_h) = \nu_b\langle\nabla g, \nabla\psi\rangle \le \nu_b|\nabla\psi|$. Write $\psi = s\varphi_h + \chi$. By Lemma 6.1, $-L(s\varphi_h) = s\lambda_F\varphi_h - s\Gamma$, so $-L\chi = X_s - \nu_b X_s$, and

$$\nu_b|\nabla\chi|^2 = \mathrm{Cov}(\chi, X_s) \le \sqrt{\mathrm{Var}\,\chi\ \mathrm{Var}\, X_s} \le \sqrt{\nu_b|\nabla\chi|^2\,\mathrm{Var}\, X_s/\gamma}$$

by Lemma 6.3. Hence $\nu_b|\nabla\chi|^2 \le \mathrm{Var}\, X_s/\gamma$, and $\nu_b|\nabla\psi| \le s\sqrt{\nu_b|\nabla\varphi_h|^2} + \sqrt{\nu_b|\nabla\chi|^2}$. □

The choice $s = 1/\lambda_F$ removes φ_h from X_s and leaves the drift alone; $s = 0$ is the plain spectral-gap bound. The best s depends on b and is found box by box.

[6] The direct Bochner bound, with Minkowski's inequality in place of the cross term, gives $(\sqrt{\mu_0} - \frac{g}{2}r)^2 + \frac{3}{2} - r$ with $g^2 = \max_W \|PA(W)\|^2 = 2.0655$; it is exact at $b = 0$, where the gap is $\frac{8}{3}$, but falls below γ for $r > 0.24$ and vanishes before $r = 1$.

6.5. **Orbits of staple sums.**

Lemma 6.5. *Let $V, W \in \mathrm{SU}(3)$.*

(i) *$L_h(VbW) = L_{V^{-1}hW^{-1}}(b)$ and $L_h(\bar b) = L_{\bar h}(b)$.*
(ii) *Every $b \in \mathbb{C}^{3\times 3}$ is in the orbit of some $\mathrm{diag}(a_1, a_2, z_3)$ with $a_1 \ge a_2 \ge |z_3|$ and $\operatorname{Im} z_3 \ge 0$.*
(iii) *A diagonal matrix lies in K if and only if it lies in the convex hull of the diagonal torus T of $\mathrm{SU}(3)$.*

Proof. (i) If U has law ν_{VbW} then WUV has law ν_b; the map preserves d and Haar measure, and $\varphi_h(U) = \varphi_{V^{-1}hW^{-1}}(WUV)$. For $\bar b$ use $U \mapsto \bar U$. (ii) Take a singular value decomposition $b = V_1 \Sigma W_1$ and move the determinants of V_1, W_1 into the last entry; a complex conjugation makes $\operatorname{Im} z_3 \ge 0$. (iii) Let c be diagonal with nonzero entries of distinct moduli and let U maximise $\operatorname{Re}\operatorname{Tr}(cU)$ on $\mathrm{SU}(3)$. The derivative along $\mathfrak{su}(3)$ vanishes, so the anti-Hermitian part of cU is a multiple $i\lambda I$ of the identity and $cU = H + i\lambda I$ with H Hermitian. Then $H^2 + \lambda^2 I = cc^*$ is diagonal with distinct entries, H commutes with it and is diagonal, and $U \in T$. Hence $\max_{\mathrm{SU}(3)} \operatorname{Re}\operatorname{Tr}(cU) = \max_T \operatorname{Re}\operatorname{Tr}(ct)$ for such c, and by continuity for every diagonal c. A diagonal matrix outside $\operatorname{conv} T$ is separated from it by such a functional, which then separates it from K. □

Since every $b \in rK$ has operator norm at most r,

$$\Lambda(r) = \sup\Big\{ \sup_{\|h\|=1} L_h(b) : b \in \mathcal{R}_r \Big\},$$

where $\mathcal{R}_r$ is the set of $\mathrm{diag}(a_1, a_2, x_3 + iy_3) \in r \operatorname{conv} T$ with $a_1 \ge a_2 \ge |x_3 + iy_3|$ and $y_3 \ge 0$, a subset of the box $[0, r]^2 \times [-r, r] \times [0, r]$ in the coordinates (a_1, a_2, x_3, y_3).

6.6. **Exact moments and interpolation.**

Lemma 6.6. *For $D, D' \in \mathbb{C}^{3\times 3}$ let $\mathbb{E}_H$ be the Haar mean on $\mathrm{SU}(3)$ and*

$$Z(D, D') = \mathbb{E}_H \exp\big(\tfrac{1}{2}\operatorname{Tr}(UD) + \tfrac{1}{2}\operatorname{Tr}(\bar U D')\big).$$

Then

$$Z(D, D') = \sum_{p \ge q \ge 0} \frac{s_{(p,q)}(DD'^{\top})\, G_{p,q}(\det D)\, G_{p,q}(\det D')}{d_{p,q}},$$

with $G_{p,q}(z) = \sum_{k\ge 0} c_{(p+k,q+k,k)} z^k$, $c_\lambda = 2^{-|\lambda|} / \prod \mathrm{hooks}(\lambda)$, $d_{p,q} = \tfrac{1}{2}(p-q+1)(q+1)(p+2)$ and $s_{(p,q)}$ the Schur polynomial. The moments of degree j in U and k in $\bar U$ are $\mathbb{E}_H[u^\alpha \bar u^\beta e^{\operatorname{Re}\operatorname{Tr}(Ub)}] = 2^{|\alpha|+|\beta|} \partial_D^\alpha \partial_{D'}^\beta Z(b, \bar b)$, the entry u_{ij} pairing with D_{ji}.

Proof. We use $(\operatorname{Tr} X)^n = \sum_{\lambda \vdash n} f^\lambda s_\lambda(X)$, where f^λ is the number of standard tableaux of shape λ, given by the hook length formula [30, Cor. 7.21.6]. Expanding the exponential gives

$$e^{\frac{1}{2}\operatorname{Tr} X} = \sum_{n \ge 0} \frac{(\operatorname{Tr} X)^n}{2^n n!} = \sum_\lambda c_\lambda s_\lambda(X),$$

the sum over partitions with at most three parts. For $U \in \mathrm{SU}(3)$ one has $s_{\lambda+(k,k,k)}(UD) = (\det D)^k s_\lambda(UD)$, so

$$e^{\frac{1}{2}\operatorname{Tr}(UD)} = \sum_{p \ge q} G_{p,q}(\det D)\, \chi_{p,q}(UD),$$

where $\chi_{p,q}$ is the irreducible character of highest weight $(p, q, 0)$ and $d_{p,q}$ its dimension. Since $\bar U = (U^{-1})^\top$, we have $\chi_{p,q}(\bar U D') = \chi_{p,q}(U^{-1} D'^{\top})$, and Schur orthogonality [11, Ch. 2],

$$\mathbb{E}_H\big[\chi_\rho(UA)\chi_{\rho'}(U^{-1}B)\big] = \delta_{\rho\rho'} \frac{\chi_\rho(AB)}{d_\rho},$$

gives the expansion. The bound below gives absolute convergence. Finally $\operatorname{Re}\operatorname{Tr}(Ub) = \tfrac{1}{2}\operatorname{Tr}(Ub) + \tfrac{1}{2}\operatorname{Tr}(\bar U \bar b)$. □

The computation keeps $p \le 24$ and $k \le 8$. Every omitted partition has more than 24 boxes, and since $|s_\lambda(X)| \le \dim(\lambda)\|X\|^{|\lambda|}$ and $\sum_{\lambda\vdash n} f^\lambda \dim\lambda = 3^n$ by Schur–Weyl duality, the omitted part of each factor is at most $\sum_{n>24}(\frac{3}{2}\|D\|)^n/n!$; Cauchy's estimate on a polydisc around b turns this into a bound for every derivative (Appendix D).

Lemma 6.7 (Interpolation on a box)**.** *Let $Q = \prod_k [l_k, l_k + w_k]$ in the coordinates (a_1, a_2, x_3, y_3), with corners b_i and multilinear weights $\omega_i(b) \ge 0$, $\sum_i \omega_i(b) = 1$, and let $\varsigma = w_1^2 + w_2^2 + \max(w_3^2, w_4^2)$. For $b \in Q$:*

(i) *$Z(b) \ge e^{-\varsigma/8} \sum_i \omega_i(b) Z(b_i)$;*
(ii) *$\nu_b(f) \le e^{\varsigma/8} \max_i \nu_{b_i}(f)$ for every $f \ge 0$ that does not depend on b;*
(iii) *if $Y_b = (\mathcal{A} + \sum_k (b_k - \beta_k)\mathcal{B}_k)^2$ with functions $\mathcal{A}, \mathcal{B}_k$ of U and a fixed point β, then*

$$\nu_b(Y_b) \le e^{\varsigma/8}\Big(\max_i \nu_{b_i}(Y_{b_i}) + \sum_k \frac{w_k^2}{4}\,\frac{\max_i Z(b_i)\nu_{b_i}(\mathcal{B}_k^2)}{\min_i Z(b_i)}\Big).$$

Proof. Write $E(b) = e^{\operatorname{Re}\operatorname{Tr}(Ub)} = \prod_k e^{c_k b_k}$, where

$$c = (\operatorname{Re} u_{11}, \operatorname{Re} u_{22}, \operatorname{Re} u_{33}, -\operatorname{Im} u_{33}),$$

so that $|c_k| \le 1$ and $c_3^2 + c_4^2 \le 1$. In each coordinate the chord of $e^{c_k x}$ over $[l_k, l_k + w_k]$ lies above the function and, by Hoeffding's lemma [19] for a two-point variable, below $e^{c_k x} e^{c_k^2 w_k^2/8}$. Multiplying,

$$E(b) \le \sum_i \omega_i(b) E(b_i) \le E(b)\, e^{\sum_k c_k^2 w_k^2/8} \le E(b)\, e^{\varsigma/8}. \tag{6.4}$$

Integrating the right inequality gives (i), and (ii) follows from the left one and (i). For (iii) let $\mathcal{F}(b) = Y_b E(b)$ and I_k the linear interpolation in the k-th coordinate, so that $\mathcal{F} - I_1 I_2 I_3 I_4 \mathcal{F} = \sum_k I_1 \cdots I_{k-1}(\mathcal{F} - I_k \mathcal{F})$. Along the k-th coordinate $\mathcal{F}$ is $\ell^2 e^{c_k x}$ times a positive factor, with ℓ affine of slope $\mathcal{B}_k$, and $\partial_k^2 \mathcal{F} = ((c_k \ell + 2\mathcal{B}_k)^2 - 2\mathcal{B}_k^2)E \ge -2\mathcal{B}_k^2 E$. With the Green kernel of the interval, whose integral is at most $w_k^2/8$,

$$\mathcal{F} - I_k \mathcal{F} \le 2\mathcal{B}_k^2 \int G\, E \le \tfrac{w_k^2}{4} \mathcal{B}_k^2 \sum_i \omega_i^{(k)} E(b_i)$$

by the left inequality of (6.4), with weights $\omega_i^{(k)} \ge 0$ of sum one that do not depend on U. The operators I_j have nonnegative weights of sum one; integrate over U and divide by $Z(b)$, bounded below by (i). □

6.7. **The cover and the theorem.** Fix a box Q and an integer j, and put $s = 3j/8192$, so that $1 - s\lambda_F = 1 - j/1024$. Since $\Gamma = \sum_k b_k G_{F_k,h}$ with $F = (e_{11}, e_{22}, e_{33}, ie_{33})$ and $G_{c,h} = \langle \nabla\varphi_c, \nabla\varphi_h \rangle$, the function X_s is affine in b. For any constant κ, $\operatorname{Var}(X_s) \le \nu_b[(X_s - \kappa)^2]$; take $\kappa = \langle \kappa(b), h \rangle$ with $\kappa(b)$ affine in b, fitted to the means at the corners. Then $X_s - \kappa = \mathcal{A} + \sum_k (b_k - \beta_k)\mathcal{B}_k$ with β the centre of Q, and Lemmas 6.4 and 6.7 give, for all $b \in Q \cap \mathcal{R}_r$,

$$\begin{aligned}\sup_h L_h(b) \le{}& s\sqrt{e^{\varsigma/8} \max_i \lambda_M(b_i)} \\ &+ \sqrt{\frac{e^{\varsigma/8}}{\gamma(r)}\Big(\max_i \lambda_V(b_i) + \frac{e_N}{\min_i Z(b_i)}\Big)},\end{aligned} \tag{6.5}$$

where $e_N = \sum_k \frac{w_k^2}{4} \max_i Z(b_i)\lambda_k(b_i)$ and λ_M, λ_V, λ_k are the largest eigenvalues of the quadratic forms $h \mapsto \nu_{b_i}|\nabla\varphi_h|^2$, $h \mapsto \nu_{b_i}[(X_s - \kappa)^2]$ at $b = b_i$, and $h \mapsto \nu_{b_i}[\mathcal{B}_k^2]$ on $\mathbb{C}^{3\times 3} = \mathbb{R}^{18}$. By Lemma 6.6 their entries are moments of ν_{b_i} of degree at most four, computed with rigorous error bounds; each largest eigenvalue is bounded above by an $LDL^\top$ factorisation of tI minus the computed form with certified positive pivots, plus the norm of its error matrix by Weyl's inequality [18, 21]. For diagonal b the measure ν_b is invariant under conjugation

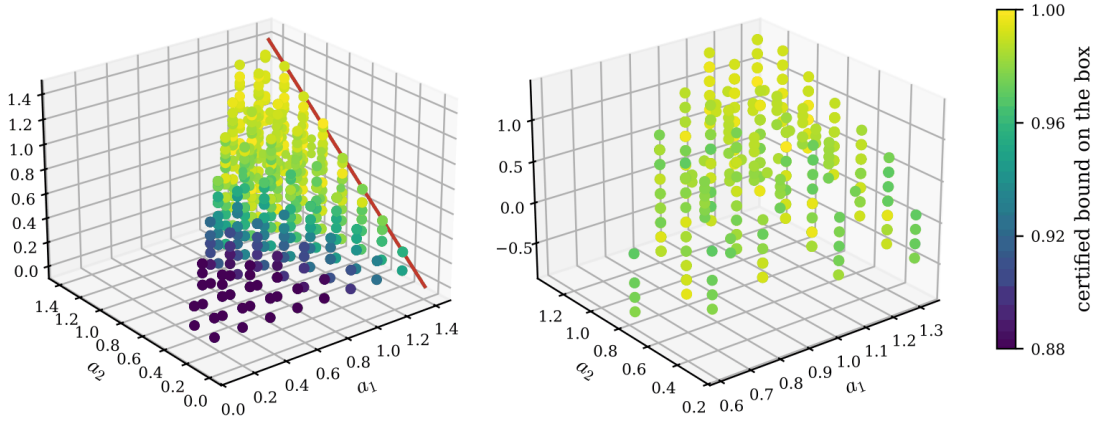


FIGURE 3. The 865 certified boxes at $r = \frac{45}{32}$, plotted at their centres in the coordinates (a_1, a_2) of the staple sum $\operatorname{diag}(a_1, a_2, x_3 + iy_3)$, coloured by the certified bound on $2r \sup_h L_h$ over the box. (a) All boxes, with height $|x_3 + iy_3|$; the red line is the face $\operatorname{diag}(r, \rho, \rho)$ on which the floating-point maximum of Lemma 6.4 lies. (b) The boxes with bound above 0.97, with height x_3; the largest bounds sit in the interior of $\mathcal{R}_r$, not at its extreme points.

by T, so each form splits into four blocks: the diagonal entries of h and the three pairs $\{h_{jk}, h_{kj}\}$.

A box is discarded when it contains no point with $a_1 \geq a_2 \geq |z_3|$, or when a linear functional $\ell(b) = \operatorname{Re}\operatorname{Tr}(c^* b)$ satisfies $\min_Q \ell > r \max_T \ell$, which separates it from $r \operatorname{conv} T$ and hence, by Lemma 6.5(iii), from $\mathcal{R}_r$.

Proposition 6.8. *$2r\Lambda(r) \leq 0.999657$ at $r = \frac{45}{32}$.*

Proof. The box $[0, r]^2 \times [-r, r] \times [0, r]$ is cut into 8192 boxes of side $r/8$. Of these, 841 survive the two tests, and 797 satisfy $2rB_Q < 1$ for the right side B_Q of (6.5), with $\gamma(r) = \frac{193}{128}$. Each of the other 44 is cut into sixteen boxes of side $r/16$; 68 of these survive the tests, and all satisfy $2rB_Q < 1$. The 865 certified boxes cover $\mathcal{R}_r$. The largest bound is 0.999657, on the box $[1.0547, 1.2305] \times [0.7031, 0.8789] \times [0.3516, 0.5273] \times [0.7031, 0.8789]$; the median is 0.966, and j ranges from 776 to 1020. Figure 3 shows the cover, and Appendix D the error analysis. □

Proof of Theorem 1.4. Proposition 6.8 gives

$$\Lambda(\tfrac{45}{32}) \leq \frac{0.999657}{45/16} < 0.355435.$$

For $\beta_W \leq \frac{45}{64}$ we have $2\beta_W \leq \frac{45}{32}$, and (6.1) with the monotonicity of Λ gives

$$\alpha \leq 4\beta_W \Lambda(\tfrac{45}{32}) < 1.42174\, \beta_W \leq 0.999657.$$

Proposition 3.2 applies with the Hilbert–Schmidt metric. □

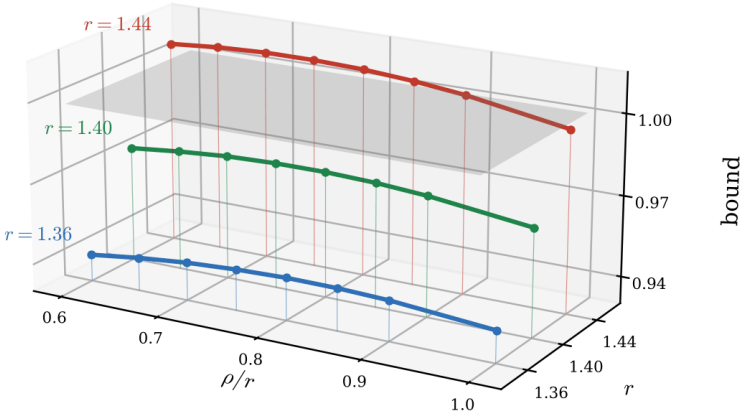


FIGURE 4. The floating-point value of $2r \sup_h$ of the bound of Lemma 6.4 on the face $b = \mathrm{diag}(r, \rho, \rho)$, as a function of ρ/r, for $r = 1.36$, 1.40 and 1.44. The grey plane is the Dobrushin threshold one. The bound crosses it between $r = 1.40$ and 1.44, where the single-link method with the gap of Lemma 6.3 ends.

7. WHERE THE METHODS STOP

SU(2). Theorem 1.2 closes the single-link route in every forest gauge at $\beta_W = \frac{1}{3}$. Going further needs blocks of several links, where the influence of a boundary link on a whole block is bounded at once; the corresponding Dobrushin–Shlosman criterion [15] is gauge invariant and stable [8, Thm. G], but its constants have not been evaluated past $\frac{1}{3}$.

Curvature. Remark 5.4 locates the end of the curvature method in this gauge between $|\beta| = \frac{1}{29}$ and $\frac{1}{26.25}$.

SU(3). The route of Section 6 bounds the $H^{-1}(\nu_b)$ norm of $\varphi_h - \nu_b \varphi_h$, which is at least $L_h(b)$, and needs it uniformly on rK. Its Galerkin values (polynomials of degree two) at matrices of rK give $2r$ times the norm equal to 0.79 at $r = 1.25$, 0.89 at $r = 1.4$, 0.97 at $r = 1.5$ and 1.009 at $r = 1.55$, so the route stops at or below $\beta_W \approx 0.77$ even with exact constants. Lemma 6.4 with Lemma 6.3 stops earlier: in floating point its largest values on $\mathcal{R}_r$ lie on the face $\mathrm{diag}(r, \rho, \rho)$ with ρ between $0.75r$ and $0.8r$, and $2r$ times the bound is 0.9458 at $r = 1.36$, 0.9767 at $r = 1.40$ and 1.0080 at $r = 1.44$ (Figure 4). Between $\beta_W \approx 0.715$ and 0.77 the room lies in the gap bound, whose Galerkin values at $r = 1.14$ are near 2.3 against $\gamma = 1.64$; past 0.77 one needs blocks.

The four open problems. Our earlier paper [8] listed four problems between the strong-coupling results and the Clay problem [20]: (1) uniqueness and a gap at intermediate fixed coupling; (2) the crossover from strong coupling to Bałaban's weak-coupling regime; (3) the continuum limit with the Osterwalder–Schrader axioms; and (4) a gap in physical units. The present paper moves the boundary of (1): for SU(2) from $\beta_W = \frac{2}{9}$ to $\frac{1}{3}$, for SU(3) from $\frac{9}{32}$ to $\frac{45}{64}$, and for every SU(N) past the curvature ceiling. It does not close (1); the region between these values and the scaling region, near $\beta_W \approx 2.2$ for SU(2) and $\beta_W \approx 6$ for SU(3), is open, and with it problem (2). Problems (3) and (4) are untouched. Every bound proved here is uniform in lattice units, so by [9] none of them can by itself produce a continuum mass gap; that requires control of the gap in physical units along the renormalisation group flow [3, 10].

Declarations

Data availability. The analytic proofs are complete as written, except the evaluation in Proposition 6.8, which is a computation with rigorous error bounds. The archive `creelie/4Y-M` on Zenodo (release v3.0.1, doi:10.5281/zenodo.23484057; all versions, doi:10.5281/zenodo.23444338) contains the programs of that computation, in Python with python-flint (FLINT and Arb), with their outputs and the list of certified boxes; independent checks in exact rational arithmetic of Lemmas 5.2 and 6.2 and of (6.3); numerical checks of Lemma 5.1 on tori; the searches of Remark 5.4 and Section 7; a Lean 4 certificate of the line field (Lemmas 2.1 and 2.2 for the table of Appendix A, by `decide`); and the source of the preprint version of this paper. Deep Bhattacharjee carried out the computations and the Lean, Python and shell checks with AI assistance from Claude (Anthropic). The first composite behind Theorem 1.1 was assembled by Anaxia, the reasoning engine of the BLLM system, in release v3.0.5 [6], as described in the footnote to that theorem; release v3.1.0 [7] records the engine's Yang–Mills runs in its failure log. Both authors approved the final text.

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Appendix A. The line field

Table 2 lists, for each residue class of x modulo two, the direction $\sigma(x)$ and the permutation $\pi_x(\mu) = \sigma(x + e_\mu)$. In every row π_x fixes $\sigma(x)$ and is a three-cycle on the other directions, which is the condition of Lemma 2.1.

Table 2. The line field: x modulo two (digits $x_0x_1x_2x_3$), the direction $\sigma(x)$ and the values $\sigma(x + e_\mu)$ for $\mu = 0, 1, 2, 3$.

x	$\sigma(x)$	$\mu = 0$	$\mu = 1$	$\mu = 2$	$\mu = 3$
0000	0	0	2	3	1
1000	0	0	3	1	2
0100	2	3	0	2	1
1100	3	2	0	1	3
0010	3	1	2	0	3
1010	1	3	1	0	2
0110	2	1	3	2	0
1110	1	2	1	3	0
0001	1	2	1	3	0
1001	2	1	3	2	0
0101	1	3	1	0	2
1101	3	1	2	0	3
0011	3	2	0	1	3
1011	2	3	0	2	1
0111	0	0	3	1	2
1111	0	0	2	3	1

Appendix B. The comparison lemma

Let (G, δ) be a compact group with a bi-invariant metric of diameter D, $\Omega = G^A$, and $C_{ee'}$ the coefficients of Section 3, with row and column sums at most $\alpha < 1$ and $C_{ee'} = 0$ unless e, e' share a plaquette, so that $|e - e'| \le 1$ for the distance of link midpoints. Fix $\lambda > 0$ with $\theta = \alpha e^{\lambda} < 1$.

Lemma B.1. *Let $S \subset A$, possibly infinite, and let μ_1, μ_2 be probability measures on Ω invariant under γ_e^F for every $e \notin S$. Let $T \subset A \setminus S$ be finite, $w_b = e^{-\lambda \operatorname{dist}(b,T)}$, and suppose $\sum_{z\in S} w_z < \infty$. There is a coupling of μ_1, μ_2 whose link distances $\varrho(b) = \mathbb{E}\,\delta(\sigma_b, \sigma'_b)$ satisfy*

$$\sum_{b\notin S} w_b \varrho(b) \le \frac{\theta D}{1-\theta} \sum_{z\in S} w_z.$$

If $S = \emptyset$ then $\mu_1 = \mu_2$.

Proof. The couplings form a weakly compact set and $\Psi(\pi) = \sum_b w_b \mathbb{E}_\pi \delta(\sigma_b, \sigma'_b)$ is weakly continuous because $\sum_b w_b < \infty$; let π^* be a minimiser. For $e \notin S$ replace the pair (σ_e, σ'_e) by a sample of an optimal coupling of $\gamma_e^F(\cdot \mid \sigma)$ and $\gamma_e^F(\cdot \mid \sigma')$, chosen measurably [31, Cor. 5.22]. Both μ_i are invariant, so the result is again a coupling, and changing the finitely many links that enter γ_e^F one at a time gives $W_1 \le \sum_{e'} C_{ee'} \delta(\sigma_{e'}, \sigma'_{e'})$. Minimality gives $\varrho(e) \le \sum_{e'} C_{ee'} \varrho(e')$. Multiply by w_e and sum over $e \notin S$; where $C_{ee'} \neq 0$, $w_e \le e^{\lambda} w_{e'}$, so

$$\sum_{e\notin S} w_e \varrho(e) \le \theta \Big[\sum_{e'\notin S} w_{e'} \varrho(e') + \sum_{z\in S} w_z \varrho(z) \Big],$$

all sums are finite, and $\varrho \le D$ on S. If $S = \emptyset$, $\varrho \equiv 0$. □

Corollary B.2. *Let μ be the Gibbs measure of γ^F, F a local function with support S_F, L_F-Lipschitz in each link, and H bounded with finite support S_H disjoint from S_F. Then*

$$|\operatorname{Cov}_\mu(F, H)| \le \|H\|_\infty L_F \frac{\theta D}{1-\theta} |S_H|\, e^{-\lambda \operatorname{dist}(S_F, S_H)}.$$

Proof. Condition on the links of S_H [22, Ch. 3]; the conditional measures are invariant under γ_e^F for $e \notin S_H$, and Lemma B.1 with $S = S_H$, $T = S_F$ bounds the difference of two of them, as in [8, Cor. 29]. □

Appendix C. The blocks of Lemma 5.2

Diagonal blocks. Let $w_a = e^{i\phi}$. On the diagonal part, $f_p = -\cos\phi\,(y_1 + y_2 - y_3 - y_4)^2$ and $c_p = 2(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)$, so

$$\begin{aligned}\mathcal{F} = \sum_i \alpha_i y_i^2 &+ \cos\phi\,(y_1 + y_2 - y_3 - y_4)^2 \\ &- \tfrac{9}{5}(y_1y_4 - y_1y_2 + y_2y_3 - y_3y_4)\end{aligned}$$

with $y_k = 0$ and weights $\alpha_{k\pm1} = \frac{21}{10}$, $\alpha_{k+2} = \frac{5}{4}$. It is affine in $\cos\phi$, so positivity at $\cos\phi = \pm1$ suffices. For $k = 1$ and $\cos\phi = 1$ the matrix in (y_2, y_3, y_4) is

$$\begin{pmatrix} \frac{31}{10} & -\frac{19}{10} & -1 \\ -\frac{19}{10} & \frac{9}{4} & \frac{19}{10} \\ -1 & \frac{19}{10} & \frac{31}{10} \end{pmatrix},$$

and Table 3 lists the leading principal minors for every position; all are positive.

Table 3. Leading principal minors of the diagonal blocks of Lemma 5.2.

Fixed link	$\cos\phi = 1$			$\cos\phi = -1$		
e_1, e_4	$\frac{31}{10}$,	$\frac{673}{200}$,	$\frac{8421}{2000}$	$\frac{11}{10}$,	$\frac{53}{200}$,	$\frac{21}{2000}$
e_2	$\frac{31}{10}$,	$\frac{861}{100}$,	$\frac{8421}{2000}$	$\frac{11}{10}$,	$\frac{21}{100}$,	$\frac{21}{2000}$
e_3	$\frac{9}{4}$,	$\frac{673}{200}$,	$\frac{8421}{2000}$	$\frac{1}{4}$,	$\frac{53}{200}$,	$\frac{21}{2000}$

Pair blocks. Let $w_a = e^{i(\chi+\vartheta)}$ and $w_b = e^{i(\chi-\vartheta)}$. The terms of f_p carry $w_a + \bar{w}_b = 2\cos\chi\, e^{i\vartheta}$ or its conjugate, and c_p carries $w_a\bar{w}_b = e^{2i\vartheta}$, so the Hermitian matrix $M(m, \vartheta)$ of $\mathcal{F}$,

$m = \cos\chi$, is affine in m with $M(-1,\vartheta) = M(1,\vartheta+\pi)$, and $m = 1$ suffices. For the fixed link e_2, in the variables (z_1, z_3, z_4) and with $c = \cos\vartheta$,

$$M = \begin{pmatrix} \frac{21}{5} + 2c & -2e^{i\vartheta} & -\frac{9}{5} - 2e^{i\vartheta} \\ -2e^{-i\vartheta} & \frac{21}{5} + 2c & \frac{9}{5} + 2e^{-i\vartheta} \\ -\frac{9}{5} - 2e^{-i\vartheta} & \frac{9}{5} + 2e^{i\vartheta} & \frac{5}{2} + 2c \end{pmatrix}.$$

With $u = \cos^2(\vartheta/2) \in [0,1]$ its leading principal minors are

$$\tfrac{20u+11}{5}, \quad \tfrac{(20u+1)(20u+21)}{25}, \quad \tfrac{400u(40u^2-23u+4)+21}{250},$$

and $40u^2 - 23u + 4$ has negative discriminant $529 - 640$. The other positions give the same determinant and the leading minors $\frac{20u+11}{5}$ or $\frac{8u+1}{2}$ and $\frac{800u^2-180u+53}{50}$, whose discriminant $32400 - 169600$ is negative. Every block is positive definite.

The inequality is not tight. In floating point the least eigenvalue of the whole form over $W \in \mathrm{U}(N)$, $N \le 4$, is 0.0149, which is also the least eigenvalue of the exact blocks; this slack is what the choice of the weights $\frac{21}{10}$, $\frac{5}{4}$, $\frac{9}{10}$ trades against the constant $\frac{29}{2}$.

Appendix D. The certified computation

Truncation. For a corner b of a box the moments of Lemma 6.6 are derivatives of Z at $(b, \bar{b})$. The truncated series is $\mathbb{E}_H[A_t(UD)B_t(\bar{U}D')]$ with A_t the part of $\sum_\lambda c_\lambda s_\lambda(UD)$ over partitions with $\lambda_1 - \lambda_3 \le 24$ and $\lambda_3 \le 8$, and B_t likewise. Every omitted partition has $|\lambda| > 24$, because $3 \cdot 9 > 24$, and the omitted part is at most $T(\|D\|) = \sum_{n>24} (\frac{3}{2}\|D\|)^n/n!$ in modulus. On the polydisc $|D_{ij} - b_{ij}| \le \rho$, $\|D\| \le \|b\| + 3\rho$, and Cauchy's estimate bounds the α-th derivative of the omitted part by $\alpha!\rho^{-|\alpha|}T(\|b\| + 3\rho)$. The derivatives of the full exponential are at most $2^{-|\alpha|}e^{v/2}$ with $v = \sum_k |b_k|$, since $|\operatorname{Tr}(Ub)| \le \sum_k |b_k|$ for diagonal b. With $\varepsilon_\alpha = 2^{|\alpha|}\alpha!\rho^{-|\alpha|}T(\|b\| + 3\rho)$ the truncated moment differs from the true one by at most $\varepsilon_\alpha e^{v/2} + (e^{v/2} + \varepsilon_\alpha)\varepsilon_\beta$, and ρ is chosen to minimise $\rho^{-4}T$.

Rounding. The truncated series is a Taylor polynomial of total degree four in the 36 real directions of $(D - b, D' - \bar{b})$, computed in double precision. Every operation is repeated on absolute values and carries an operation count n, so that the computed coefficient differs from the exact one by at most $\gamma_n = nu/(1 - nu)$, $u = 2^{-53}$, times the absolute-value coefficient [18, Section 3.1]. The polynomials φ_h, $\langle \nabla\varphi_h, \nabla\varphi_{h'} \rangle$ and $G_{F_k,h}$ have exact Gaussian rational coefficients, given by Lemma 6.1:

$$\begin{aligned} G_{c,h}(U) = {} & \tfrac{1}{2}\operatorname{Re}\operatorname{Tr}(ch^*) - \tfrac{1}{2}\operatorname{Re}\operatorname{Tr}(cUhU) \\ & - \tfrac{1}{3}\operatorname{Im}\operatorname{Tr}(cU)\operatorname{Im}\operatorname{Tr}(hU). \end{aligned}$$

Normalisation by Z, the quadratic forms and their error matrices are formed with the same bookkeeping, and the largest eigenvalues are bounded in ball arithmetic [21]. The off-block entries of the exact forms vanish and are dropped with their error bounds checked.

Separation. The functional c of a discarded box is the dual solution of a linear program for the gauge of the centre of the box with respect to a 72×72 grid of T. Only the final inequality is used: $\max_T \ell$ is bounded by its maximum over a 512×512 grid plus $\frac{1}{2}H\delta^2/2$, with $\delta = 2\pi/512$ and $H = |c_1| + |c_2| + 2|c_3|$ a bound for its Hessian on the torus, and the inequality is checked with relative margin 10^{-12}.

Choice of s. The integer j is chosen per box by a floating-point search; any choice gives a valid bound. The cover runs in about half an hour on four cores and writes every certified box with its j and its bound.

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